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Problem-based learning and student outcomes in higher education: a second-order meta-analysis

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ABSTRACT

Despite a substantial corpus of first-order meta-analysis studies examining the impact of problem-based learning (PBL) model on student outcomes, no previous study has yet attempted to synthesize the findings of these studies. The objective of this study is to investigate the impact of PBL on a range of student outcomes in higher education, including critical thinking, theoretical knowledge, clinical skills, student attitude, and student satisfaction. To this end, a statistical analysis was conducted on 20 first-order meta-analyses obtained from electronic databases. In this study, a second-order meta-analysis was employed to synthesize the findings of the first-order meta-analysis studies. The statistical analyses were conducted using the random-effects model. The study concluded that PBL has a significant and high impact on student outcomes. The mean effect size is ES = .60 [CI = 0.49; 0.71; Q(t) = 757.75]. Furthermore, the analysis revealed that the moderators influencing the variability of the effects of PBL encompass the type of student outcome, geographical location, sampling methodology employed in the primary studies, research quality, publication year, and publication type.

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Problem-based learning; higher education; second-order meta-analysis; student outcomes; critical thinking; clinical skills; student attitude

Introduction

In the twenty-first century, employees are required to possess the capacity to overcome novel obstacles and engage in complex communication to fulfill their roles effectively (Koenig 2011). This is due to the advent of the new century, which brought with it a new economic order and thus the necessity for the acquisition of new skills, including problem-solving (Funke, Fischer, and Holt 2018; Noss 2012). In particular, the accelerated pace of development in science and technology has rendered problem-solving a critical skill in all aspects of life (Hidayat, Susilaningsih, and Kurniawan 2018). Cınsequently, higher education institutions have been compelled to adapt their curricula to reflect this shift. The anticipated behavioral patterns of students in higher education are becoming increasingly diverse to align with the objectives of a knowledge-based society. These behaviors are specifically oriented toward the resolution of authentic problems and encompass advanced cognitive abilities, including problem-solving and critical or analytical thinking. To foster these higher-order cognitive skills in students, problem-based learning (PBL) has been introduced in tertiary education. PBL is increasingly regarded as a valuable approach for fostering critical, analytical, and creative problem-solving skills in students at higher education institutions (Hallinger 2020), and it aligns with recent higher education policy trends, including push for competency-based education,

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employability, and lifelong learning. The implementation of PBL practices, which originated in Canada, has subsequently been adopted in the United States, Europe, and eventually in Asia (Chan et al. 2022). Since the 1960s, higher education institutions have integrated PBL into their curricula, including those offering degrees in medicine, nursing, pharmacy, dentistry, and engineering (Azer 2017).

The changes in the pedagogical realm also facilitated the emergence of PBL. In the latter half of the twentieth century, there was a notable shift in educational practices as educators came to recognize the limitations of teaching methods that were grounded in behaviorist and cognitivist theories. Consequently, they commenced the investigation of novel models that extend beyond mere knowledge acquisition (Hung, Jonassen, and Liu 2008). The construction of new skills necessitated the implementation of novel pedagogical approaches that integrated content knowledge with skill acquisition. PBL, a constructivist model of teaching, proved an efficacious remedy for this need (Erdem 2024). PBL provides an enhanced learning environment through the utilization of scenarios based on authentic, real-world problems, enabling students to develop their problem-solving abilities, enhance their critical thinking skills and expand their content knowledge (Gustin et al. 2018; Moallem 2019; Savery 2006; Strobel and Van Barneveld 2009). Besides, the foundations and practices of PBL are not confined to the constructivist model of teaching, but it is in assocation with various learning theories and cognition including andragogy learning, transformative learning, experiential learning, social learning, information processing, collaborative/cooperative learning, contextual learning, cognitive load, cognitive learning, and discovery learning (Gewurtz et al. 2016).

The aforementioned advantages have resulted in PBL being adopted in a wide range of contexts across the globe, and a substantial corpus of research has accumulated on it. Literature on PBL has more than doubled after 2010, indicating a growing interest in the subject (Hallinger 2021). As a result of this accumulation, systematic evaluations have been conducted using qualitative data analysis methods to synthesize the findings of primary research on PBL (Anggraeni et al. 2023; Li et al. 2019; Wosinski et al. 2018). Similarly, researchers have endeavored to review and synthesize the findings regarding the effects of PBL at the K-12 level (Erdem 2024; Jensen 2015) or postsecondary education (Liu and Pásztor 2022; Sayyah et al. 2017; Zhang et al. 2015) using a meta-analysis approach. Notwithstanding the review studies that have been conducted, the existing literature continues to debate the relative merits and disadvantages of PBL compared to traditional lecture-based instruction. This ongoing discussion can be attributed to discrepancies in the manner in which PBL is implemented and the particular contexts in which it is employed. The diverse application of problem-based learning across disciplines, differences in age groups and content areas, unclear definitions of learning concepts and outcomes, varying implementation methods, and the diversity of problems employed during the learning process all contribute to this uncertainty (Moallem 2019). In light of the pivotal role of PBL in higher education, it is imperative to conduct second-order meta-analysis studies to elucidate the remaining uncertainties surrounding the impact of PBL on a diverse range of student outcomes. This is because some existing first-order meta-analysis (FOM) studies have been limited to a single student outcome, such as critical thinking (Kong et al. 2014; Liu and Pásztor 2022). Some studies have focused exclusively on the impact of PBL on students' academic achievement, as evidenced by Jensen (2015), Sugano and Nabua (2020), and Wilder (2015). However, Hallinger (2021) indicated a shift in topical focus of recent PBL research, from academic performance to other student outcomes such as critical thinking and attitudes.

In their 2009 study, Strobel and Van Barneveld synthesized qualitative findings of research on FOM studies spanning higher education and other levels. It is also essential to synthesize the quantitative findings of FOM research. FOM studies represent data from various contexts. Consequently, second-order meta-analyses, which include multiple outcomes based on FOM, may provide a comprehensive overview of the literature and facilitate a deeper understanding of the phenomenon. This study aims to conduct a thorough investigation into the impact of problem-based learning on various student outcomes in higher education, including theoretical knowledge, clinical skills, critical thinking, student attitudes, and student satisfaction, as provided in the first-order meta-analysis data set of the present study. This study provides a more comprehensive and up-to-date understanding of the effectiveness of PBL in higher education. To this end, the objective of this study is to integrate the findings of first-order meta-analysis studies through the utilization of a second-order meta-analysis approach. Consequently, a comparative analysis can be conducted to ascertain the impact of PBL on student outcomes in relation to various moderators, including country, student outcome, type of higher education program, and characteristics of the studies. In alignment with the aforementioned objective, this study sought to address the following:

1. What is the impact level of PBL on student outcomes in higher education?
2. Does the impact of PBL on student outcomes in higher education vary according to moderator variables?

Literature review

Problem-based learning

Problem-based learning (PBL) was initially developed for medical education in the mid-1960s and has since been applied across various disciplines worldwide (Barrows 2000; Hung, Jonassen, and Liu 2008; Schmidt, Vermeulen, and van der Molen 2006; Torp and Sage 2002). PBL is an experience-based approach that fosters learning through the investigation, explanation, and resolution of meaningful problems (Barrows 2000; Savery 2015; Schmidt, Vermeulen, and van der Molen 2006; Torp and Sage 2002). This approach is noteworthy for situating students at the core of the learning process, wherein problems serve as the point of departure (Lin et al. 2010). In particular, students engage with complex problems in small groups, commencing their work without any initial instruction on the subject matter (Barrows 1986; Hallinger 2021). Consequently, the role of the teacher is transformed into that of a facilitator who guides the learning process rather than simply providing knowledge (Hmelo-Silver 2004). This shift in roles has a profound impact on the dynamics of both students and teachers, fostering a more interactive and self-driven learning experience (Hallinger 2021).

In the context of the PBL process, students are presented with realistic and engaging problem scenarios. Such scenarios prompt students to engage in discourse concerning their extant knowledge and to identify their deficiencies (Loyens et al. 2015). Subsequently, they analyze the information, identify deficiencies, and address these gaps through a self-directed learning (SDL) process. As students gain new knowledge, they test their hypotheses and evaluate the learning outcomes, thereby consolidating their understanding when the problem is resolved (Begay et al. 2006; Hmelo-Silver 2004). Consequently, PBL facilitates students in attaining a more profound comprehension and more enduring retention of information over an extended period (Van Blankenstein et al. 2011). Furthermore, it facilitates the development of critical thinking and reflective learning abilities, particularly when confronted with intricate challenges (Dochy et al. 2003). In addition to fostering these skills, PBL emphasizes collaborative learning and prepares students for lifelong learning (Hmelo-Silver 2004). This demonstrates that PBL is not merely a cyclical process but is also firmly anchored to significant learning theories. Indeed, PBL is firmly grounded in constructivist learning theories, which posit that students actively construct knowledge through self-directed learning and group interactions (Hmelo-Silver 2004; Loyens et al. 2015; Savery 2006). PBL also relies on the use of meta-cognition, which involves goal-setting, strategy selection, and goal evaluation. In PBL, students need to use their meta-cognitive skills of monitoring one's own learning process, which includes analyzing the problem and and aspects of problem solving, in this case (Gijselaers 1996).

PBL is rooted in constructivism as an approach to learning and teaching, but it is also related to the principles of various other theories of learning and cognition. Gewurtz et al. (2016) refer to adult/andragogy learning, transformative/emancipatory learning, experiential learning, social learning, vicarious learning, information processing, collaborative/cooperative learning, contextual learning, cognitive load, cognitive learning, and discovery learning as associated with PBL. Predominantly used in higher education, PBL covers all these established and new theories of learning and cognition through incorporating principles such as that learning should be independent, self-directed, goal-oriented, internally motivated, applicable to practice, involve cognitive processes, require active engagement, provide interaction between learners, and activate prior knowledge (Gewurtz et al. 2016; Hessler and Henderson 2013; Lin 2013; Roberts 2010). Furthermore, it is a more logical and effective approach in higher education, particularly in medical schools, as it fosters the development of complex problem-solving and self-directed learning competencies. In higher education, students' advanced self-regulation skills and flexible course structures facilitate the implementation of PBL in these environments (Hmelo-Silver 2004). Consequently, it is crucial to investigate its application and impacts within the context of higher education.

PBL in higher education

Given the provenance of PBL in the medical field, it is unsurprising that it is also a favorable approach in higher education. It is an effective pedagogical approach across a range of disciplines, notably engineering, medical education, and entrepreneurship. The utilization of PBL in higher education has become pervasive, particularly in disciplines such as engineering (Perrenet, Bouhuijs, and Smits 2000), medical science (Dolmans et al. 2005; Kilroy 2004), and entrepreneurship (Sousa and Costa 2022). A body of research has indicated that PBL facilitates the development of problem-solving abilities and the generation of more effective solutions to complex engineering problems (Perrenet, Bouhuijs, and Smits 2000). In the context of medical education, PBL has been demonstrated to facilitate the development of critical thinking, collaboration, and self-management skills, contribute to students' deep learning processes, and enhance the retention of knowledge (Dolmans et al. 2005; Hmelo-Silver 2004; Wirkala and Kuhn 2011). Furthermore, in the context of entrepreneurship education, PBL has been demonstrated to enhance students' entrepreneurial abilities, including innovative thinking and risk-taking (Sousa and Costa 2022).

PBL aligns with recent trends in higher education policy, including competency-based education (Anderson 2018), employability (Rögens, Scoupe, and Beausaert 2020), and lifelong learning (Yang, Schneller, and Roche 2015). In the contemporary higher education landscape, an increasing demand exists for institutions to cater to a more diverse student population, including a growing number of adult learners. In response to this demand, educational institutions are exploring innovative approaches to meet these needs, with competency-based education (CBE) emerging as a prominent solution. The introduction of CBE programs provides a cost-effective pathway to degree completion, allowing students to progress at their own pace, guided by a mentor or coach (Anderson 2018). PBL aligns with this approach by prioritizing active learning, problem solving and critical thinking, ensuring that students develop and demonstrate competencies in real-world contexts. CBE requires student-centered and self-directed learning, and PBL practices have been shown to produce fruitful learning outcomes in CBE settings (Sistermans 2020).

The purpose of higher education is not only to facilitate the acquisition of knowledge but also to enable students to apply their skills to a diverse workplace to meet the employment requirements that are in demand today (Lurie and Garrett 2017). PBL serves this purpose as it enhances problem-solving, teamwork, adaptability, communication, and self-directed learning. The utilization of PBL has been demonstrated to enhance the employability skills of university students, thereby equipping them with the capacity to respond positively to evolving circumstances and novel challenges (Clausen and Andersson 2019) as well as their absorptive capacity (Li et al. 2020). Additionally, lifelong learning policies encourage the development of self-regulated learners who can continuously acquire and apply new knowledge throughout their careers. PBL fosters lifelong learning skills by encouraging self-directed inquiry, metacognition, and reflective thinking. The utilization of PBL in higher education contexts has demonstrated enhanced lifelong learning skills among students (Alt and Raichel 2022; Dunlap 2005; Hernandez-Encuentra and Sanchez-Carbonell 2005).

PBL facilitates the development of twenty-first-century skills, particularly creative thinking, teamwork, and effective communication (Moreno 2018; Thorndahl and Stentoft 2020). It is generally accepted that PBL practices encourage student-centered learning in higher education, facilitate the transformation of teaching methods (Birch 1986; Margetson 1994), and have a positive impact on both academic and personal development (Ibragimov, Ibragimova, and Bakulina 2016). These findings provide compelling evidence of the efficacy of PBL across diverse academic fields in higher education and its role in fostering students' cognitive, affective, and academic success. In addition to its effectiveness across a range of disciplines, its benefits for students in higher education are significant. Therefore, it is essential to concentrate on the impact of PBL on student outcomes in higher education.

The impact of PBL on the outcomes of tertiary students

As a pedagogical approach, PBL has been extensively examined in the context of higher education, with findings indicating that it can have a considerable impact on student outcomes. The research evidence indicates that PBL has a positive impact on both academic achievement and student development across a range of domains, including theoretical knowledge, clinical skills, critical thinking, student attitudes, and satisfaction (Hmelo-Silver 2004). The extensive scope of these effects indicates that PBL offers students a more profound learning experience. This study examines the effects of PBL on a variety of student outcomes, as investigated in first-order meta-analysis studies included in the dataset of this study. The outcomes encompass theoretical knowledge, clinical skills, critical thinking, student attitudes, and student satisfaction.

Theoretical knowledge

The term 'theoretical knowledge' is used to describe an understanding of the fundamental concepts, principles, and theories that underpin a particular discipline. The impact of the PBL model on theoretical knowledge in higher education is evidenced by a range of empirical findings. The evidence suggests that PBL has a positive impact on the acquisition of theoretical knowledge by students. The approach encourages students to engage actively with and comprehend information through the analysis of real-world problems, which can facilitate a more profound and efficacious grasp of theoretical concepts (Ma and Lu 2019; Xu, Ye, and Wang 2021). Nevertheless, in certain instances, the impact on knowledge acquisition is comparable to that of traditional teaching methods. This variability is frequently attributed to specific characteristics of the educational context, suggesting that the efficacy of PBL may not be universally superior (Wang et al. 2016). In light of these findings, the impact of PBL on theoretical knowledge in higher education is context-dependent, and may yield disparate results in disparate applications and educational environments.

Clinical skills

PBL has been identified as an effective learning model for developing students' clinical skills in higher education. The development of clinical skills necessitates the acquisition of practical knowledge through analysis of real patient scenarios. The impact of PBL on the advancement of such skills has been a subject of considerable emphasis, particularly within the context of pediatric medical education (Ma and Lu 2019). Students who have undergone training in PBL have demonstrated notable advancement in their clinical abilities, benefiting from the practical applications and problem-solving processes that it offers. It has been demonstrated to have a beneficial impact on students' clinical decision-making processes, enabling them to make more rapid and effective decisions in a range of clinical scenarios (Ma and Lu 2019; Xu, Ye, and Wang 2021). Furthermore, students who have undergone training in this method display enhanced confidence and competence at the outset of their professional careers. Furthermore, PBL has been demonstrated to positively influence the long-term development of clinical skills, with students' practical competencies becoming more permanent (Zhang et al. 2015). These findings underscore the pivotal role of PBL in the acquisition and advancement of clinical competencies in health education.

Critical thinking

The term 'critical thinking' is defined as a set of skills and habits related to the organization, analysis, evaluation, and ultimately the decision-making process regarding one's own cognitive processes. Such abilities encompass the ability to approach information critically, to question, and evaluate different perspectives (Ellerton 2020; Willingham 2008). Ellerton (2022) examined the role of critical thinking in the interpretation and utilization of information. The potential of PBL to develop critical thinking skills in higher education has been investigated, and the ways in which students acquire and apply these skills have been revealed. The role of PBL in enhancing critical thinking is of great importance for understanding the effectiveness of teaching methods and how they influence students' approach to information.

Ellerton (2022) posits that the PBL model is efficacious in developing students' critical thinking abilities, engendering more profound thinking processes, and facilitating students' utilization of critical thinking skills in a productive manner. PBL is particularly effective in developing students' capacity to solve complex problems and to think critically about these problems. Furthermore, it strengthens students' abilities to analyze, evaluate and solve problems. In particular, the opportunity to address real-world problems afforded by PBL supports critical thinking throughout the process (Kumar and Refaei 2017). Furthermore, PBL enhances students' capacity to evaluate diverse perspectives and problem-solving abilities (Saunders-Stewart, Gyles, and Shore 2012). It encourages students to learn through the problems they encounter, thereby facilitating the development of their critical thinking skills throughout the process. The active research, analysis, and synthesis of information during problem-solving reinforces critical thinking skills (Ellerton 2022). A recent meta-analysis also confirmed the strong effect of PBL on tertiary students' critical thinking skills (Liu and Pásztor 2022). In conclusion, PBL assists students in progressing beyond mere superficial knowledge, thereby facilitating a more profound comprehension and critical assessment of information.

Student attitudes

Student attitude can be defined as the general feelings and thoughts that students develop toward a specific subject, learning method, or educational environment (Akhtar 2007). PBL has been demonstrated to facilitate the development of a more positive attitude toward scientific subjects, enhance interest in science-related topics, and cultivate a more enthusiastic and positive approach to scientific subjects through active participation in problem-solving processes (Ferreira and Trudel 2012; Shin and Kim 2013). It has a more favorable impact on students' attitudes toward their courses than more traditional teaching methods. Conversely, the extent of the impact of PBL may be contingent on several variables (Brice 2017; Demirel and Dağyar 2016).

Student satisfaction

PBL has been demonstrated to engender positive changes in students' attitudes, facilitating a more constructive approach to the learning process, and enhancing overall satisfaction (Shin and Kim 2013). Furthermore, it has the potential to stimulate students' intrinsic motivation, enhance their learning capabilities, and cultivate deeper enthusiasm for learning (Li et al. 2024). In this context, examining the impact of PBL on student satisfaction is crucial. The extant literature indicates that PBL enhances student satisfaction and the learning experience in higher education (Shin and Kim 2013). It has been demonstrated that this approach increases students' satisfaction and positively affects their academic performance (Mejias et al. 2015). Furthermore, the model has been found to result in higher levels of satisfaction and interest among students, increased excitement for learning, and a significant boost in satisfaction (Li et al. 2024). Consequently, PBL has been demonstrated to have a positive impact on student satisfaction and academic success (Emerald et al. 2013; Hagi and Al-Shawwa 2011; Khan et al. 2007).

Method

This study utilizes the second-order meta-analysis method as its research approach, which is employed to integrate the results of FOM studies (Oh 2020). It is utilized to combine the statistical data of meta-analysis studies conducted on the same topic or problem (Schmidt and Oh 2013). Systematic evaluations and analyses were conducted according to the PRISMA guidelines (Moher et al. 2009). The flow of information through different phases of the review in Line with PRISMA framework is provided in Figure 1.

Data collection

Data for this study were obtained from multiple electronic databases, including Scopus, Web of Science, MEDLINE, ERIC, Academic Search Ultimate, Google Scholar, and ProQuest. The database searches used specific keywords such as 'problem based' or 'PBL,' combined with terms like 'meta-analysis' or 'meta analytic' or 'systematic review'. In accordance with the study's objectives, various criteria for inclusion and exclusion were established, which are detailed below.

Inclusion and exclusion criteria

1) FOM studies must be articles published in English language until the year 2024.
2) FOM studies should include experimental studies focusing on PBL and student outcomes. FOM without a control group of traditional teaching methods are excluded. FOM studies related to PBL supported by web tools, computers, teamwork, or other methods (i.e. enriched PBL) are excluded.

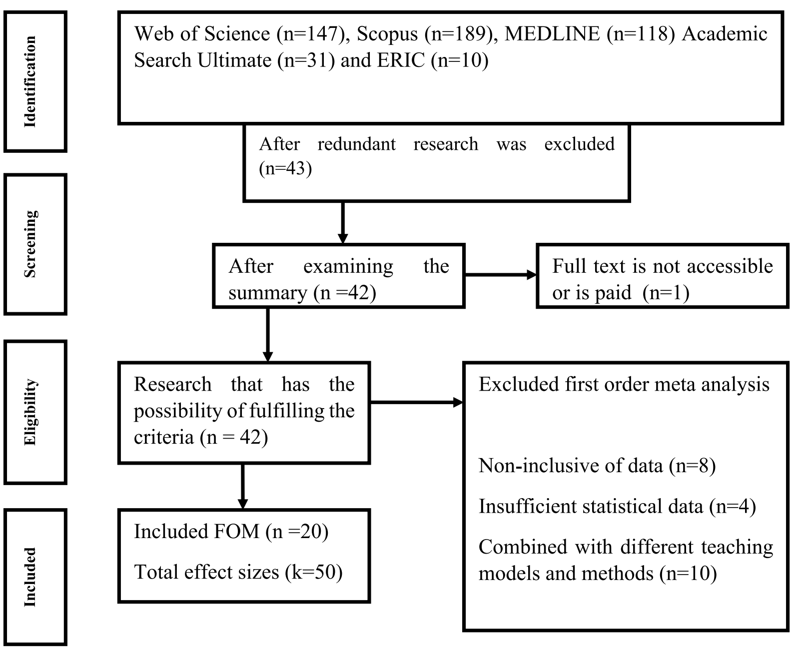

Figure 1. Data flow diagram.

Similarly, if an FOM study encompasses project-based learning, case-based learning, or other different learning models alongside PBL, it is excluded. Such meta-analysis studies involve the effects of different methods beyond PBL, thereby causing difficulties in evaluating effect sizes. Therefore, they are excluded.

3) FOM studies must contain sufficient statistical data, such as standardized mean differences (SMD) or Cohen's d, Hedge's g, Odds Ratio, and Fisher's z, as well as parameters related to these effect sizes, such as upper limit (UP), lower limit (LL), standard error (SE), and variance value. Studies lacking sufficient statistical data to calculate effect sizes are excluded.
4) If the overlap ratio among FOM studies falls below 25%, it is presumed that the meta-analysis studies are unrelated to each other (Cooper and Koenka 2012).
5) FOM studies should cover higher education institutions. FOM studies conducted at the K12 level are excluded because they fall outside the scope of this research.
6) The control group in FOM studies should consist of traditional teaching methods.

The titles and abstracts of the studies retrieved from the database were reviewed. Following this assessment, we compiled a list of potential studies meeting the inclusion criteria to form a data pool for this study. After a detailed examination of the data pool, the dataset of this study consisted of 20 FOM studies. The database search yielded 495 studies, which diminished to 43 FOM studies after eliminating repeating studies. Various steps, as shown in the data flow diagram, were followed. The 20 FOM studies included in this study comprised 469 independent studies. The FOM studies included at least five independent studies and at most 82 independent studies. The mean of the independent studies included in FOM studies was 23.45 and its standard deviation was 18.49. The data flow diagram is presented in Figure 1. The characteristics of the meta-analysis studies comprising the dataset are summarized in Table 1.

Coding

The coding form, which portrays the attributes of FOM studies, was developed using MS Excel. The coding was performed by two researchers. Cohen's Kappa reliability coefficient was calculated as $\kappa = 0.89$. The researchers discussed codes with discrepancies. Expert opinion was sought for codes in whcih consensus could not be reached, and the final decision was made. The coding is presented in Table 2.

Quality assessment

Kung et al. (2010) revised R-AMSTAR scale to evaluate the quality of meta-analysis studies. The R-AMSTAR scale comprises 11 sections and was originally designed for use in the medical domain. Sections 8A and 8B of R-AMSTAR cover clinical applications. Therefore, sections 8A (1 point) and 8B (1 point) were not scored. The value ranges proposed by Young (2017) were used as references for assessing the R-AMSTAR scale. The quality scores of FOM studies were used as a moderator variable, and they were statistically tested. In addition, the FOM studies were coded by two researchers, and the purpose, statistical model, and selected effect sizes of the FOM studies were evaluated. After creating the dataset, the statistical procedures performed were as follows. Statistical analyses were performed using the CMA.3.0 software.

Statistical analysis

The unit of analysis in this study is the independent effect sizes within the FOM studies. The random-effects model was chosen as the statistical model. Average effect size, publication bias, moderator analysis, and heterogeneity analyses were conducted using the random-effects model.

Effect size selection: Standardized mean differences (SMD) or Cohen's d and Hedge's g effect size calculation methods yield identical outcomes in sizable samples. In other words, the gg value in

Table 1. FOM characteries.

Study	ES	LL	UL	k	Scope	R	Location	Sampling in FOM	Outcome	Quality	Year range
Xu, Ye, and Wang (2021)	.89	.52	1.26	9	Cell biology	A	Not spesific	RCTs and non-RCTs	Theoretical knowledge	Medium	NA-2021
Gao, et al. (2020)***	1.51	.79	2.23	9	Psychology	A	China	RCTs and non-RCTs	Combined	Medium	NA-2020
Ma and Lu (2019)	1.16	.79	1.52	12	Pediatri	A	China	RCTs	Theoretical knowledge	High	NA-2018
	1.56	.87	2.25	5					Clinical skills		
Sayyah et al. (2017)	.9	.56	1.24	10	Nursing	A	Iran	RCTs and non-RCTs	Combined	High	1980–2016
	.64	.6	1.13	5	Medical						
	.66	.25	1.08	4	Dental						
Wang et al. 2016	.76	.33	1.19	13	Clinical medicine	A	China	RCTs and non-RCTs	Theoretical knowledge	Medium	2007–2014
	1.46	.89	2.02	10					Clinical skills		
Zhang et al. (2018)	.82	.28	1.36	4	Radiology	A	China	RCTs and non-RCTs	Theoretical knowledge	Medium	NA-2017
	1.56	.86	2,27	5				RCTs	Clinical skills		
Kong et al. 2014	.33	.3	.52	9	Nursing	A	Not spesific	RCTs	Critical thinking	High	1965–2012
Galvao et al. (2014)a	.21	.08	.35	5	Pharmacy	A	Not spesific	RCTs and non-RCTs	Theoretical knowledge	Medium	1995–2010
	.26	.03	.49	5					Theoretical knowledge		
Zhou et al. (2016)	1.17	.77	1.57	16	Pharmacy	A	China Iran, South Korea,	RCTs and non-RCTs	Theoretical knowledge	High	1965–2014
Sharma et al. (2023)	.44	.4	.73	8	Nursing	A	China and Taiwan Iran, South Korea	RCTs	Critical Thinking Problem solving	High	2004–2021
	.25	−.02	.52	3							
Sharma et al. (2023)	.31	−.02	.65	3					Self-confidence		
Liu and Pásztor (2022)	.58	.41	.75	42	Medical	A	Not spesific	RCTs and non-RCTs	Critical thinking	High	2000–2021
	.74	.38	1.10	10	Sciences				Critical thinking		
	.91	.48	1.33	6	Arts				Critical thinking		
	.49	.14	.85	16	Medical				Critical thinking dis.		
	.87	.22	1.51	6	Sciences				Critical thinking dis.		
	.71	.42	.99	5	Arts				Critical thinking dis.		
Shin and Kim (2013)	1.32	.72	1.93	10	Nursing	A	Not spesific	RCTs and non-RCTs	Theoretical knowledge	High	1972–2012
	.92	.68	1.17	8					Clinical skills		
	1.44	1.07	1.81	19					Student satisfaction		
	.67	.39	.94	21					Learning attitude		
Zheng et al. (2023)	−.19	−.71	.33	6	Surgical	A	Not spesific	RCTs and non-RCTs	Theoretical knowledge	Medium	−2022
	.81	.12	1.49	7					Clinical competence		

	.92	.32		1.53	8					Student satisfaction		
	.26	−.37		.89	7					Comprehensive scores		
Li et al. (2024)	1.10	.78		1.41	42	Orthopedics	A	Not spesific	RCTs	Theoretical knowledge	High	Before 2023
*	2.07	1.61		2,53	31					Procedural skill		
	1.20	.88		1.52	12					Clinical skil		
**	4,70	3,20		6.93	10					Teaching interest		
**	5,43	3.83		7.69	16					Student satisfaction		
Bi et al. (2021)*	3.17	2.28		4.07	17	Obstetrics and Gynecology	A	China	RCTs	Theoretical knowledge	High	2015–2020
*	2.17	1.63		2,71	7					Clinical practice		
	1.15	.93		1.37	5					Clinical operation		
Zhang et al. (2015)	.76	.65		.88	10	Medical	A	China	RCTs and non-RCTs	Theoretical knowledge	High	Before 2014
Wei et al. (2023)	.47	.33		.61	19	Nursing	A	Iran, South Korea, China, and Turkey	RCTs and non-RCTs	Critical Thinking	Medium	2004–2021
Huang et al. (2013)	.88	.46		1.31	8	Dental	A	China	RCTs	Theoretical knowledge	Medium	2006–2012
Brice (2017)a	−.15		SE = .29		9	Medical	D	Not spesific	RCTs and non-RCTs	Attitude	Medium	2003–2016
	−.39		SE = .23		9					Theoretical knowledge		
	.02	−.02		.06	33	Medical	D	Not spesific	RCTs and non-RCTs	Theoretical knowledge	Medium	1977–2002
	.32	.27		.38	29					Clinical skills		
	.19	.03		.34	10					Problem solving		
	.47	.39		.55	19					Self-directed learning		
	.45	.39		.52	30					Attitude		

RCTs = Randomized controlled trials; non-RCTs = Non-randimized controlled trials; A = Article; D = Dissertation; * = Outlier value;** OR; *** = Hedge’s g; dis:disposition; SE = Standard Error.

Table 2. Encodings.

Group	Code
Study	Researcher(s) (year of publication)
Outcomes	Students' scores were coded exactly as found in the FOM studies and were not merged. The sudent outcomes coded are theoretical knowledge, clinical skills, critical thinking and student satisfaction. Some FOM studies reported different student outcomes by combining them. In this case, the student outcome was coded as 'combined'. If a student outcome was insufficient in number to form a group, it was coded as 'others' if k < 3. Student outcomes other than skills and knowledge, such as disposition and interest, were coded as student attitude.
Primary research location	If the FOM surveys cover a country or region, they were coded according to the location covered. If FOM studies were not limited to surveys in any specific location, they were coded as 'not specific location'.
Higher education program	Higher education programs were coded as medicine, dentistry, nursing, science and arts.
Primary research sampling method	If the FOM included primary studies with only randomized control groups, they were coded as (RCG); if both randomized and non-randomized control groups were used, they were coded (RCG and non-RCG).
FOM quality	The revised version of the Assessment of Multiple Systematic Reviews (R-AMSTAR) scale, which was revised by Kung and colleagues in 2010, was coded as insufficient, low, medium, and high.
Year range	FOM studies were coded as before-2010, 2011–2020 and after 2021.
Report type	FOM studies were coded as articles and dissertations.

smaller samples represents the adjusted value of dd (Goulet-Pelletier and Cousineau 2018; Marfo and Okyere 2019). It was assumed in this study that the meta-analysis studies included samples of sufficient size. Second-order meta-analyses are conducted using the same assumptions (Hew et al. 2021; Young 2017). The majority of studies comprising the dataset of this study reported SMD values as k = 47 effect sizes; k = 2 reported Odds Ratios (OR); and k = 1 reported Hedge's g values. The OR values were converted to SMD values, which were used in the analyses.

Heterogeneity and moderator analysis: This involved the use of Q statistics to assess the level of heterogeneity within the dataset. Q(t) indicated total heterogeneity, whereas Q(b) determined heterogeneity between groups. The variability in effect sizes across groups was examined using Q(b) values. This study employed outcome variables, educational level, primary research report type, location, and meta-analysis quality variables as moderators.

Outlier analysis: Outlier analysis was conducted for the effect sizes in the dataset, and k = 3 effect sizes were identified as outliers and not included in the statistical analyses. Thus, the analyses in this study were conducted to encompass k = 47 effect sizes. The effect sizes identified as outliers were reported along with the characteristics of the studies.

Publication bias analysis: Publication bias analyses have strengths and weaknesses compared to each other (Jin, Zhou, and He 2015). Taking this into account, publication bias analyses for the dataset included checking and interpreting the results of funnel plot, Egger's regression test, Begg and Mazumdar rank correlation, and Duval & Tweedie trim and fill analysis techniques.

Findings

Descriptive statistics, mean effect size, and publication bias

The dataset comprised k = 47 effect sizes ranging from ES = 1.15 to ES = −.39. The mean effect size was calculated as ES = .68 [CI = .57; .79; $p < .001$]. The total heterogeneity of the dataset was Q(46) = 730.67; $p < .001$, with a heterogeneity level of I2 = 93.70. The reliability of the calculated (observed) effect size was linked to publication bias. The Begg and Mazumdar rank correlation test results did not suggest publication bias (tau = −.01; z = .12; $p = .89$). This result demonstrated that there was no statitistically significant difference between the effect sizes and the number of effect sizes. However, Egger's regression test results indicated publication bias (t (45) = 5.53; $p < .001$), meaning that small effect sizes were not adequately represented in the dataset. Additionally, the Duval & Tweedie trim and fill (DTtf) test indicated publication bias for the dataset, and recommended the addition of k = 6 effect sizes to the left of the mean effect size. Adding these six studies with negative and small effect

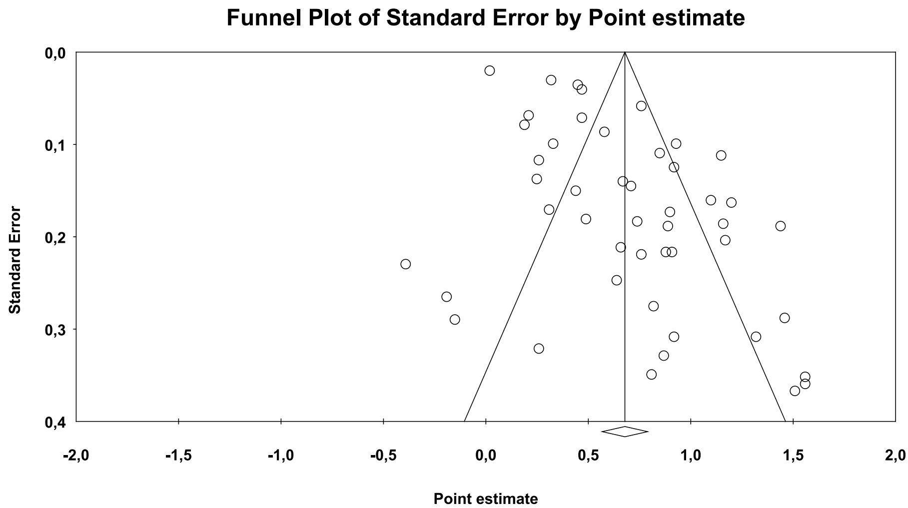

Figure 2. Funnel plot.

sizes provided a more symmetric distribution. A similar finding was also observed in the Funnel plot. After this addition, the adjusted effect size was calculated as adjusted ES = .60 [CI = 0.49; .71; Q(t) = 757.75]. The difference between the observed and adjusted effect sizes was approximately ΔES = .08, indicating a slight publication bias.

Considering the collective data, we can infer that the dataset exhibits high heterogeneity and a slight publication bias. Moreover, based on the adjusted and observed effect size values, we conclude that PBL significantly influences student outcomes. The distribution of effect sizes based on standard errors is depicted in the funnel plot graph in Figure 2.

Moderator and heterogeneity analyses

The effect size distribution indicates a high-level heterogeneity. This may be due to the characteristics of the FOM studies in the dataset. Therefore, moderator analyses were performed to explain the reasons of heterogeneity resulting from the FOM studies, and the analysis results were interpreted. The high heterogeneity may stem from the various sample groups employed in FOM studies. The samples cover different countries and higher education programs. Countries may have different approaches to education in their higher education systems, and the objectives of different higher education programs may focus on different student outcomes and competencies. These situations may have been the source of heterogeneity.

Table 3 presents moderator and heterogeneity analyses for the dataset. The effect of PBL on student outcomes varied significantly according to the outcome type (Q (6) = 14.65; p = .02). The effect of PBL on clinical skills (ES = 1.06 [CI = .75; 1.37]) and student satisfaction (ES = 1.10 [CI = .62; 1.59]) is very high, while the effect on theoretical knowledge (ES = .59 [CI = .37; .81]), critical thinking (ES = .56 [CI = .24; .89]), and student attitude (ES = .57 [CI = .26; .88]) is high. Effect sizes varied significantly according to the location covered by the FOM studies (Q (2) = 19.03; p < .01). Studies covering China (ES = 1.10 [CI = .88; 1.32]) produced very high effect sizes, while studies covering Iran (ES = .74 [CI = .34; 1.15]) and studies not specific to a location (ES = .55 [CI = .43; .66]) produced high effect sizes. Effect sizes varied statistically significantly according to the sampling types of the FOM studies (Q (1) = 4.20; p = .04). Studies with RCG sampling type (ES = .84 [CI = .65; 1.05]) produced higher effect sizes compared to studies with both RCG and non-RCG sampling types (ES = .61 [CI = .48; .73]). Effect sizes varied significantly according to the qualities of the FOM studies (Q (1) = 15.25; p < .01). Studies in the high-quality group (ES = .82 [CI = .70; .93]) produced very high effect

Table 3. Moderator and heterogeneity analysis of the dataset.

Group	k	ES	LL	UL	Q(b)	df (Q)	*p*
Outcomes					14.65	6	.02
Theoretical knowledge	15	.59	.37	.81			
Clinical skills	8	1.06	.75	1.37			
Critical thinking	6	.56	.24	.89			
Student attitude	7	.57	.26	.88			
Student satisfaction	3	1.10	.62	1.59			
Combined	4	.88	.44	1.33			
Other	4	.31	−.08	.70			
Program					1.17	5	.95
Medical	28	.68	.54	.83			
Nursing	10	.67	.44	.90			
Pharmacy	3	.50	.09	.91			
Dental	2	.77	.21	1.33			
Sciences	2	.79	.21	1.38			
Arts	2	.80	.27	1.34			
Location					19.03	2	<.01
China	11	1.10	.88	1.32			
Iran	3	.74	.34	1.15			
Not spesific	33	.55	.43	.66			
Primary research sampling method			4.20	1	.04		
RCG and non − RCG	34	.61	.48	.73			
RCG	13	.84	.65	1.04			
FOM quality			15.25	1	<.01		
High	26	.82	.70	.93			
Medium	21	.46	.32	.59			
Report					23.36	1	<.01
Article	40	.77	.67	.87			
Dissertation	7	.19	−.02	.40			
Year range					11.42	2	<.01
Before 2010	5	.29	.05	.53			
2011–2020	22	.77	.63	.91			
After 2021	20	.70	.55	.84			

sizes, while studies in the moderate-quality group (ES = .46 [CI = .32; .59]) produced moderate effect sizes. Effect sizes varied significantly according to the report type of the FOM studies (Q (1) = 23.36; $p < .01$). Studies in article form (ES = .77 [CI = .67; .87]) produced high effect sizes, while studies in dissertation form (ES = .19 [CI = −.02; .40]) produced low effect sizes. Effect sizes varied significantly according to the publication year ranges of the FOM studies (Q (2) = 11.42; $p < .01$). Studies before 2010 (ES = .29 [CI = .05; .53]) produced moderate effect sizes, while studies between 2011 and 2020 (ES = .77 [CI = .63; .91]) and studies after 2021 (ES = .70 [CI = .55; .84]) produced high effect sizes. On the other hand, effect sizes did not vary significantly according to the type of higher education program.

Discussion

This study employed the second-order meta-analysis method to examine the impact of problem-based learning (PBL) in higher education on student outcomes based on FOM studies. The analysis encompasses 20 FOM studies covering 469 indepent studies, including published articles and unpublished doctoral theses, up to 2024. The roles of outcome types, the study location, higher education program type, and various characteristics of FOM studies are examined as moderators.

Based on the extant meta-analyses, this study demonstrated that PBL has a high overall impact on student outcomes in higher education. Strobel and Van Barneveld (2009), who synthesized the qualitative data of FOM research related to PBL, support these findings as well as other review research (Dochy et al. 2003; Gijbels et al. 2005). They reported that compared to traditional approaches, PBL was more effective for long-term retention of knowledge, skill development, and student satisfaction in the learning process. In addition, the present study revealed PBL's high levels of effect on student outcomes, including theoretical knowledge, clinical skills, critical thinking, student attitude, and satisfaction. The results reveal that PBL has a very-high effect on tertiary students' clinical skills and satisfaction levels and a high effect on their theoretical knowledge, critical thinking, and attitude. Although we should be cautious in generalizing the findings, these results confirm the effectiveness of PBL on student outcomes in higher education, offering a venue for its application in tertiary degree programs.

The significant impact of PBL on students' clinical skills is particularly noteworthy in this study. Wosinski et al. (2018) also emphasized a similar result. They concluded that PBL facilitated nursing students' clinical reasoning. This may be attributed to PBL's focus on generating solutions to real-life problems, as clinical skills are directly related to problems encountered in clinical settings. Despite PBL having the highest impact on clinical skills in the present study, some earlier research reported little effect of PBL on clinical skills (Albenese 2001), which may be related to challenges in implementation of the method or quality of the instructors' abilities. On the other hand, as the current study revealed, more recent studies yielded higher effect sizes. During the course of time, better delivery of the model may be enabled, and better measurement instruments for clinical skills have been available.

As with clinical skills, PBL had a markedly very high impact on students' satisfaction with the learning environment. Furthermore, PBL had a significant impact on students' attitudes toward learning. Hallinger (2020) posits that this is due to PBL encouraging students to engage in active participation in learning activities. The affective dimensions of learning and schooling warrant greater attention (Jackson 2018). The attitudes of students toward the school, learning, and instructors are affected by the learning environment, indicating a close relationship between the learning environment and affective performance (Cheng 1994). PBL offers students the opportunity to participate actively in a problem-solving process, which has been shown to foster positive attitudes and enthusiasm (Ferreira and Trudel 2012; Shin and Kim 2013). This, in turn, has been found to result in enhanced intrinsic motivation and learning capabilities (Li et al. 2024). Furthermore, this favorable learning environment and enhanced student outcomes facilitate increased student satisfaction in higher education (Emerald et al. 2013; Li et al. 2024; Mejias et al. 2015; Shin and Kim 2013). A comparison of the effects of PBL and traditional teaching approaches on students' learning satisfaction, learning attitude, and motivation also demonstrates that PBL is more effective than traditional teaching in higher education settings (Song 2008).

The findings of this study indicate that PBL has a significant impact on the acquisition of theoretical knowledge by students. Consequently, PBL is an effective method for imparting theoretical knowledge to students. However, PBL was sometimes approached with a degree of caution, on the grounds that it was thought to be less effective than traditional approaches in the acquisition of theoretical knowledge (Albenese and Mitchell 1993; Hmelo-Silver 2004). Despite some research indicating lower levels of knowledge acquisition in PBL practices, in relation to cognitive load in PBL, evidence suggests that students taught through PBL demonstrate superior knowledge retention. Similarly, Dochy et al. (2003) reported that students in PBL demonstrated lower acquisition of theoretical knowledge but exhibited superior long-term retention of the acquired knowledge. In the same vein, Strobel and Van Barneveld (2009) concluded that students engaged in PBL demonstrated suboptimal performance in short-term retention while exhibiting superior long-term retention and performance improvement. PBL appears to facilitate deep learning, as opposed to surface learning (Dolmans et al. 2016), as students engage actively with and comprehend information through the analysis of real-world problems. This can facilitate a more profound and efficacious grasp of theoretical concepts (Ma and Lu 2019; Xu, Ye, and Wang 2021). These outcomes align with the objectives of instruction in a degree program, given that graduates will be practitioners in the field of their degree program. In that, graduates of PBL medical schools scored higher on professional competencies compared to graduates of conventional medical schools (Schmidt, Vermeulen, and van der Molen 2006).

The findings indicated that PBL had a high impact on students' critical thinking abilities. The problem-solving nature of the PBL learning environment enables students to engage in profound thinking, evaluate diverse perspectives, analyze and synthesize information, solve real-world problems and, as a result, utilize critical thinking in a productive manner, thereby developing their critical thinking skills (Ellerton 2022; Saunders-Stewart, Gyles, and Shore 2012). Indeed, critical thinking can be defined as a process of problem-solving (Paul and Elder 2003). PBL encourages students to move beyond superficial knowledge and engage in critical assessment of information, subsequently utilizing it for problem-solving purposes. The high-level effect of PBL on students' critical thinking in the current study is supported by the theoretical relation between critical thinking ability and PBL. The design of PBL supports students' critical thinking development; however, long-term exposure may be needed (Masek and Yamin 2011). Some research indicating a non-significant effect of PBL on critical thinking may be affected by the short-term nature of the intervention, as critical thinking development requires time and a number of variables. Nevertheless, PBL provides an optimal setting for the cultivation of this aptitude. Another issue is that researchers do not agree on what constitutes critical thinking in relation to PBL, which has resulted in a range of mixed findings (Thorndahl and Stentoft 2020).

In addition to examining the influence of PBL on student outputs, this study also sought to ascertain how the impact of the PBL model on student outcomes differed according to moderator variables. The initial variable investigated was the type of location. This phenomenon is referred to as location bias in meta-analysis studies (Higgins and Green 2011). The magnitude of treatment effects may vary by country in clinical trials (Vickers et al. 1998). In this study, the effect sizes observed in China were significantly higher than those observed in other locations. This may be attributed to the fact that the PBL model has been implemented in higher education institutions in China since the 1970s (Chan et al. 2022; Xu, Ye, and Wang 2021). In other words, this may be attributed to the extensive experience of Chinese higher education institutions with the PBL learning model. Similarly, a review study reported that PBL use is not universally widespread due to the need for greater human resources and continuous training (Trullas et al. 2022). Moreover, the role of Chinese culture in this finding should not be overlooked. PBL fosters independent and self-directed learning, a quality that is evident in Chinese-origin students who have been observed to possess a strong inner drive for achievement and high self-imposed career expectations (Gwee 2008). The significance of work ethics, respect toward learning, group harmony, cooperation due to collectivism, need for achievement due to competitive examinations, and resilience in Chinese culture may be effective in explaining this result.

The effect sizes observed in the FOM research differed according to the specific type of report under consideration. It is notaworthy that unpublished doctoral dissertations, or in other words, unpublished first-order meta-analyses, tend to produce lower effect sizes. While Hew et al.'s (2021) research, which covers FOMs in experimental design, falsifies this result, Tan, Gao, and Shi (2022) research, which covers FOMs in survey design, supports this result. This phenomenon may be attributed to the reluctance of journal editors to publish studies that are statistically non-significant or demonstrate low effects (Page et al. 2021; Pigott et al. 2013).

The study additionally determined that the efficacy of the PBL model is contingent on the publication year range of FOM research. The diminished effect sizes observed prior to 2010 may be attributed to the lack of experience among practitioners in conducting experimental research related to PBL. Conversely, the comparable effect sizes observed between studies conducted between 2011 and 2020 and those conducted subsequently support this proposition. Furthermore, Hallinger (2020) and Azer (2017) highlighted the evolution of the PBL model in the 2010s. In this case, researchers can monitor the effectiveness of the PBL model over time.

This study demonstrated that the inclusion of only randomized control groups (RCG) in first-order meta-analyses resulted in the production of higher effect sizes. The findings are supported by Hew et al. (2021) and Liu and Pásztor (2022), while Shin and Kim (2013) present an opposing viewpoint. This discrepancy may be attributed to the fact that primary research utilizing RCG facilitated the control of numerous variables throughout the PBL implementation process. Although conducting fully controlled experiments in the social sciences is challenging, this approach is important for elucidating the efficacy of PBL. In experimental designs where control is not sufficiently established, there is a high probability that the intervention tool's effect may be influenced by different variables (Bonell et al. 2018). Furthermore, high-quality first-order meta-analysis research has been shown to yield higher effect sizes. Similarly, Young (2017) and Martin et al. (2022) reached comparable conclusions regarding the quality of FOMs in second-order meta-analysis research.

Limitations and further research

The present study is confined to a comparative analysis of PBL and traditional learning models, as represented in FOM studies. It would be beneficial to examine the impact of PBL that has been enriched with different teaching methods and techniques on student outcomes. This study is limited to the higher education level. A further area for investigation would be the impact of PBL on student outcomes at the K-12 level. A comparison can be made between higher education and K-12 level. Moreover, this research is constrained to FOM studies conducted in English. It would be beneficial for future research to be conducted in conjunction with other languages. Conversely, the present study encompasses a restricted number of FOMs of higher education programs outside health sciences. Further studies may be conducted on FOMs covering engineering, teacher education, law, social sciences, humanities, and other higher education programs. In addition, this study is based on quantitative data. Further research can synthesize the findings of systematic evaluations of qualitative data on PBL. This study has a slight publication bias. The positive and substantial effect sizes reported in the FOM studies included in this study may be a contributing factor to the observed publication bias. Therefore, the findings should be interpreted in light of the publication bias reported in the findings section. Finally, this study reports high and very high effects of PBL on student outcomes but it should be noted that medical and nursing programs are represented with more effect sizes, whereas those from other programs are less represented. It is important to exercise caution when generalizing this situation to all higher education programs. There is a need for research to determine the effectiveness of PBL in other higher education programs.

The moderator analysis in this study also revealed significant differences in student outputs, which provide a foundation for further research. First, the FOM research indicated a location bias, with higher effect sizes observed in China. To reduce the impact of location bias, future FOM research should incorporate studies from a range of geographical locations and a variety of databases. In addition, in the event that meta-analysis studies have been conducted with a particular emphasis on a specific country or location, the findings of such studies can be integrated using second-order meta-analysis techniques. Second, the unpublished FOM research yielded comparatively lower effect sizes. It is therefore imperative that first-order meta-analysis studies be conducted which encompass unpublished master's and doctoral theses. Third, FOM research conducted prior to 2010 yielded comparatively lower effect sizes. Qualitative research examining primary research on PBL in depth can identify differences over time. This enables the identification of differences over time. Fourth, studies employing only RCTs yielded higher effect sizes. It may be beneficial for researchers to explore the possibility of conducting fully controlled interventions in order to elucidate the efficacy of PBL, although this may prove to be a challenging endeavor in social sciences. Ultimately, high-quality first-order meta-analyses yielded higher effect sizes. It is incumbent upon researchers to endeavor to adhere to the highest standards of quality in their meta-analysis.

Conclusion

Despite the theoretical and empirical support for the PBL approach, there have been some debates regarding its efficiency. Critics contend that PBL is a demanding and time-intensive approach, often implemented by individuals who may not fully grasp its complexities, and the limited evidence linking PBL to improved clinical competence raises doubts about the practicality and effectiveness of such intensive learning methods in routine professional practice (Kilroy 2004). Some challenges in PBL include crowded classes, vicious circle due to locality in problem finding, being not suitable for every student, and lack of teacher competencies (Ramadhani, Huda, and Umam 2019). The supporters and opponents of the cognitive foundation of the PBL approach continue to debate its merits. Nevertheless, evidence indicates that graduates of PBL curricula exhibit comparable or enhanced professional competencies relative to graduates from more conventional curricula (Neville 2008). Furthermore, the impact of PBL on student outcomes extends beyond academic performance.

The results of this second-order meta-analysis study demonstrate a significant overall impact of PBL on student outcomes. The present study, which is based on a large sample that has been represented in FOM studies, offers evidence of the significant impact of PBL on student outcomes. However, it is important to exercise caution when generalizing these findings, considering the publication bias and high level of heterogeneity. This indicates that the model is efficacious not only with regard to academic performance but also with respect to practice-based outputs, such as clinical skills and critical thinking, and affective outputs, such as student attitude and student satisfaction. It can be concluded that PBL should be employed in higher education settings to facilitate enhanced theoretical knowledge acquisition, augmented clinical and critical thinking skills, and positive student attitudes and increased satisfaction. In-service training sessions on PBL practices could be organized for faculty members experiencing difficulties in maintaining student satisfaction and positive attitudes. In particular, if the field of study is related to clinical skills, PBL appears to be the optimal model of instruction for the program. The findings of this study corroborate the efficacy of PBL in relation to the aforementioned student outcomes, thereby substantiating the existing literature. Nevertheless, further research is undoubtedly necessary to ascertain the full potential of PBL (Stentoft 2017).

Disclosure statement

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